Chapter Summary

Key Terms

10.1 Points, Lines, and Planes

- line segment
- plane
- union
- intersection
- parallel
- perpendicular

10.2 Angles

- vertex
- right angle
- acute angle
- obtuse angle
- straight angle
- complementary
- supplementary

10.3 Triangles

- acute
- obtuse
- isosceles
- equilateral
- hypotenuse
- congruence
- similarity
- scaling factor

10.4 Polygons, Perimeter, and Circumference

- perimeter
- polygon
- pentagon
- hexagon
- heptagon
- octagon
- quadrilateral
- trapezoid
- parallelogram
- circumference

10.5 Tessellations

- tessellation
- translation
- reflection
- rotation
- glide reflection

10.6 Area

- triangle
- square
- rectangle
- rhombus
- apothem
- radius
- circle

10.7 Volume and Surface Area

- surface area
- volume
- right prism
- right cylinder

10.8 Right Triangle Trigonometry

- right triangle
- sine
- cosine
- tangent

Key Concepts

10.1 Points, Lines, and Planes

- Modern-day geometry began in approximately 300 BCE with Euclid's *Elements*, where he defined the principles associated with the line, the point, and the plane.
- Parallel lines have the same slope. Perpendicular lines have slopes that are negative reciprocals of each other.
- The union of two sets, A and B, contains all points that are in both sets and is symbolized as $A \cup B$.
- The intersection of two sets A and B includes only the points common to both sets and is symbolized as $A \cap B$.

10.2 Angles

- Angles are classified as acute if they measure less than $90°$, obtuse if they measure greater than $90°$ and less than $180°$, right if they measure exactly $90°$, and straight if they measure exactly $180°$.
- If the sum of angles equals $90°$, they are complimentary angles. If the sum of angles equals $180°$, they are supplementary.
- A transversal crossing two parallel lines form a series of equal angles: alternate interior angles, alternate exterior angles, vertical angles, and corresponding angles

10.3 Triangles

- The sum of the interior angles of a triangle equals $180°$.
- Two triangles are congruent when the corresponding angles have the same measure and the corresponding side lengths are equal.
- The congruence theorems include the following: SAS, two sides and the included angle of one triangle are congruent to the same in a second triangle; ASA, two angles and the included side of one triangle are congruent to the same in a second triangle; SSS, all three side lengths of one triangle are congruent to the same in a second triangle; AAS, two angles and the non-included side of one triangle are congruent to the same in a second triangle.
- Two shapes are similar when the proportions between corresponding angles, sides or features of two shapes are equal, regardless of size.

10.4 Polygons, Perimeter, and Circumference

- Regular polygons are closed, two-dimensional figures that have equal side lengths. They are named for the number of their sides.
- The perimeter of a polygon is the measure of the outline of the shape. We determine a shape's perimeter by calculating the sum of the lengths of its sides.
- The sum of the interior angles of a regular polygon with n sides is found using the formula $S = (n - 2)\,180°$. The measure of a single interior angle of a regular polygon with n sides is determined using the formula $a = \frac{(n-2)180°}{n}$.
- The sum of the exterior angles of a regular polygon is $360°$. The measure of a single exterior angle of a regular polygon with n sides is found using the formula $b = \frac{360°}{n}$.
- The circumference of a circle is $C = 2\pi r$, where r is the radius, or $C = \pi d$, and d is the diameter.

10.5 Tessellations

- A tessellation is a particular pattern composed of shapes, usually polygons, that repeat and cover the plane with no gaps or overlaps.
- Properties of tessellations include rigid motions of the shapes called transformations. Transformations refer to translations, rotations, reflections, and glide reflections. Shapes are transformed in such a way to create a pattern.

10.6 Area

- The area A of a triangle is found with the formula $A = \frac{1}{2}bh$, where b is the base and h is the height.
- The area of a parallelogram is found using the formula $A = bh$, where b is the base and h is the height.
- The area of a rectangle is found using the formula $A = lw$, where l is the length and w is the width.
- The area of a trapezoid is found using the formula $A = \frac{1}{2}h\left(b_1 + b_2\right)$, where h is the height, b_1 is the length of one base, and b_2 is the length of the other base.
- The area of a rhombus is found using the formula $A = \frac{d_1 d_2}{2}$, where d_1 is the length of one diagonal and d_2 is the length of the other diagonal.
- The area of a regular polygon is found using the formula $A = \frac{1}{2}ap$, where a is the apothem and p is the perimeter.
- The area of a circle is found using the formula $A = \pi r^2$, where r is the radius.

10.7 Volume and Surface Area

- A right prism is a three-dimensional object that has a regular polygonal face and congruent base such that that lateral sides form a $90°$ angle with the base and top. The surface area SA of a right prism is found using the formula $SA = 2B + ph$, where B is the area of the base, p is the perimeter of the base, and h is the height. The volume V of a right prism is found using the formula $V = Bh$, where B is the area of the base and h is the height.
- A right cylinder is a three-dimensional object with a circle as the top and a congruent circle is the base, and the side forms a $90°$ angle to the base and top. The surface area of a right cylinder is found using the formula $SA = 2\pi r^2 + 2\pi rh$, where r is the radius and h is the height. The volume is found using the formula $V = \pi r^2 h$, where r is the radius and h is the height.

10.8 Right Triangle Trigonometry

- The Pythagorean Theorem is applied to right triangles and is used to find the measure of the legs and the hypotenuse according the formula $a^2 + b^2 = c^2$, where c is the hypotenuse.
- To find the measure of the sides of a *special* angle, such as a $30°$-$60°$-$90°$ triangle, use the ratio $x : x\sqrt{3} : 2x$, where each of the three sides is associated with the opposite angle and $2x$ is associated with the hypotenuse, opposite the $90°$ angle.
- To find the measure of the sides of the second *special* triangle, the $45°$-$45°$-$90°$ triangle, use the ratio $x : x : x\sqrt{2}$, where each of the three sides is associated with the opposite angle and $x\sqrt{2}$ is associated with the hypotenuse, opposite the $90°$ angle.
- The primary trigonometric functions are $\sin\theta = \frac{opp}{hyp}$, $\cos\theta = \frac{adj}{hyp}$, and $\tan\theta = \frac{opp}{adj}$.
- Trigonometric functions can be used to find either the length of a side or the measure of an angle in a right triangle, and in applications such as the angle of elevation or the angle of depression formed using right triangles.

Videos

10.5 Tessellations

- M.C. Escher: How to Create a Tessellation (https://openstax.org/r/create_tessellation)
- The Mathematical Art of M.C. Escher (https://openstax.org/r/mathematical_art)

Formula Review

10.2 Angles

To translate an angle measured in degrees to radians, multiply by $\frac{\pi}{180}$.

To translate an angle measured in radians to degrees, multiply by $\frac{180}{\pi}$.

10.4 Polygons, Perimeter, and Circumference

The formula for the perimeter P of a rectangle is $P = 2L + 2W$, twice the length L plus twice the width W.

The sum of the interior angles of a polygon with n sides is given by

$$S = (n - 2)180°.$$

The measure of each interior angle of a regular polygon with n sides is given by

$$a = \frac{(n - 2)180°}{n}.$$

To find the measure of an exterior angle of a regular polygon with n sides we use the formula

$$b = \frac{360°}{n}.$$

The circumference of a circle is found using the formula $C = \pi d$, where d is the diameter of the circle, or $C = 2\pi r$, where r is the radius.

10.6 Area

The area of a triangle is given as $A = \frac{1}{2}bh$, where b represents the base and h represents the height.

The formula for the area of a square is $A = s \cdot s$ or $A = s^2$.

The area of a rectangle is given as $A = lw$.

The area of a parallelogram is $A = bh$.

The formula for the area of a trapezoid is given as $A = \frac{1}{2}h\left(a + b\right)$.

The area of a rhombus is found using one of these formulas:

- $A = \frac{d_1 d_2}{2}$, where d_1 and d_2 are the diagonals.
- $A = \frac{1}{2}bh$, where b is the base and h is the height.

The area of a regular polygon is found with the formula $A = \frac{1}{2}ap$, where a is the apothem and p is the perimeter.

The area of a circle is given as $A = \pi r^2$, where r is the radius.

10.7 Volume and Surface Area

The formula for the surface area of a right prism is equal to twice the area of the base plus the perimeter of the base times the height, $SA = 2B + ph$, where B is equal to the area of the base and top, p is the perimeter of the base, and h is the height.

The formula for the volume of a rectangular prism, given in cubic units, is equal to the area of the base times the height, $V = B \cdot h$, where B is the area of the base and h is the height.

The surface area of a right cylinder is given as $SA = 2\pi r^2 + 2\pi rh$.

The volume of a right cylinder is given as $V = \pi r^2 h$.

10.8 Right Triangle Trigonometry

The Pythagorean Theorem states

$$a^2 + b^2 = c^2$$

where a and b are two sides (legs) of a right triangle and c is the hypotenuse.

Projects

1. One of the reasons so many formulas in geometry were discovered was because of the importance in finding measurements of lengths, areas, perimeter, and angles. Find at least five examples of how geometry can be used in practical applications today.
2. Who were the Pythagoreans? Why did this society exist? Explore what they did and discuss some of their beliefs.

Chapter Review

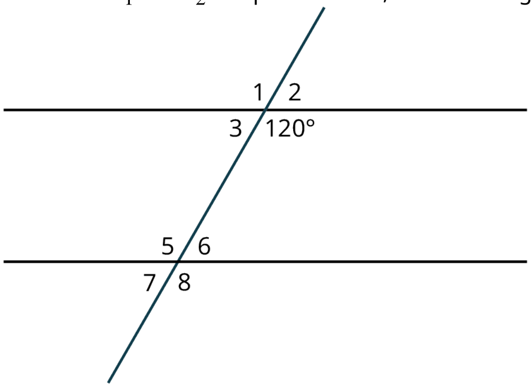
Points, Lines, and Planes

Use the figure shown for the following exercises.

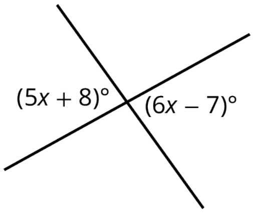
1. Find $\overrightarrow{BC} \cup \overrightarrow{CD}$.
2. Find $\overline{AC} \cap \overline{BC}$.
3. Name the points in the set $\overrightarrow{BD} \cap \overleftarrow{CA}$.

Angles

4. Given that l_1 and l_2 are parallel lines, solve the angle measurements for all the angles in the figure shown.

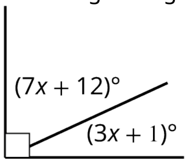


5. Find the measure of the vertical angles in the figure shown.

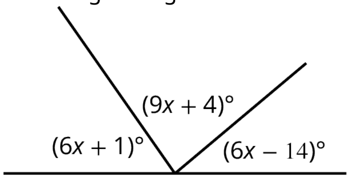


6. Classify the angle as acute, right, or obtuse: $79°$.

7. Use the given figure to solve for the angles.

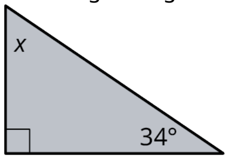


8. Use the given figure to solve for the angle measurements.

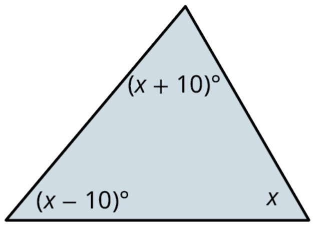


Triangles

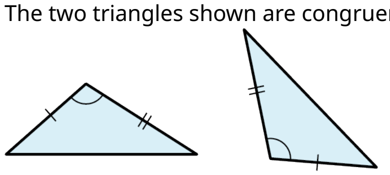
9. Use the given figure to find the measure of the unknown angles.



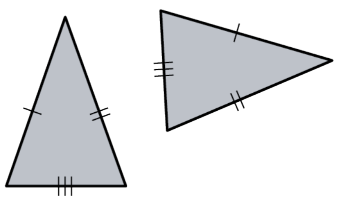
10. Use algebra to find the measure of the angles in the figure shown.



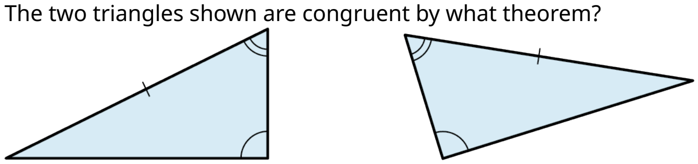
11. The two triangles shown are congruent by what theorem?



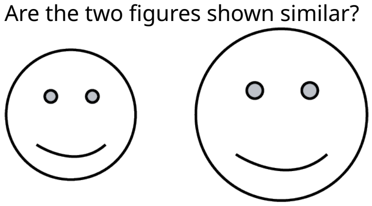
12. The two triangles shown are congruent by what theorem?



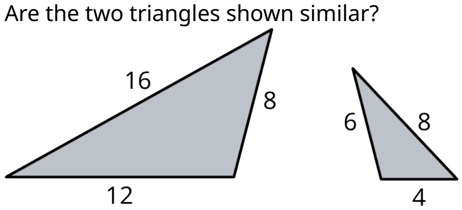
13. The two triangles shown are congruent by what theorem?



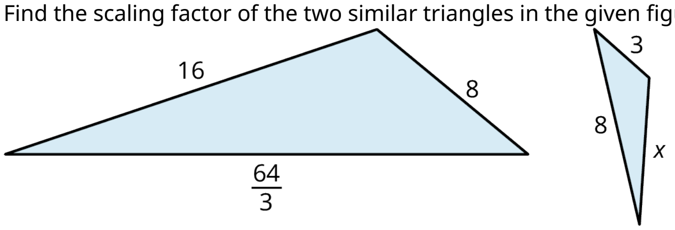
14. Are the two figures shown similar?



15. Are the two triangles shown similar?



16. Find the scaling factor of the two similar triangles in the given figure.

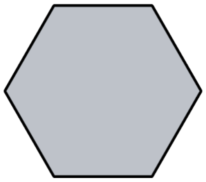


17. Find the length of x in the figure shown.

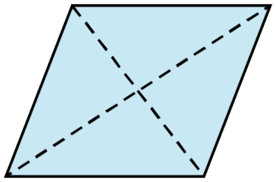
[Polygons, Perimeter, and Circumference]

Identify the polygons.

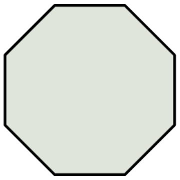
18.



19.

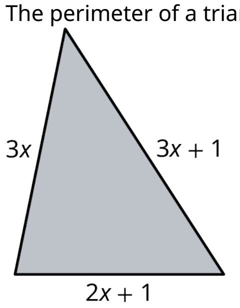


20.



21. Find the perimeter of a regular hexagon with side length 5 cm.

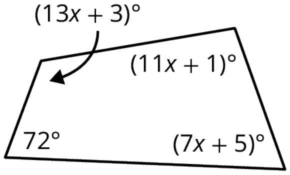
22. The perimeter of a triangle is 18 in the given figure. Find the length of the sides.



23. What is the measure of an interior angle of a regular heptagon?

24. What is the measure of an interior angle of a regular octagon?

25. Calculate the measure of each interior angle in the figure shown.



26. Find the measure of an exterior angle of a regular pentagon.

27. What is the sum of the measures of the exterior angles of a regular heptagon?

28. Find the circumference of a circle with radius 3.2 cm.

29. Find the diameter of a circle with a circumference of 35.6 in.

Tessellations

30. In what field of mathematics does the topic of tessellations belong?

31. Tessellations can have no _______or _________.

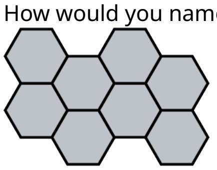
32. What type of transformation moves an object over horizontally by some number of units?

33. What type of transformation results in a mirror image of the shape?

34. Which regular polygons will tessellate the plane by themselves?

35. The sum of the interior angles of the shapes meeting at a vertex is equal to how many degrees?

36. How would you name this tessellation?

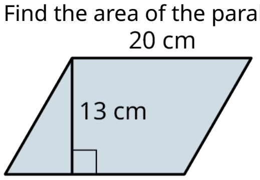


Area

37. What is the area of a triangle with a base equal to 5 cm and a height equal to 12 cm?

38. If the area of a triangle equals $24\ \text{cm}^2$ and the base equals 8, what is the height?

39. Find the area of the parallelogram shown.

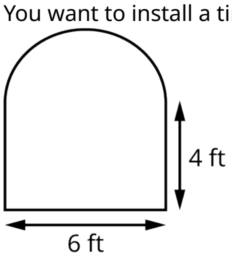


40. If the area of a trapezoid equals $\frac{45}{2}\ \text{cm}^2$, $b_1 = 3\ \text{cm}$, and the height equals $h = 5\ \text{cm}$, find the length of b_2.

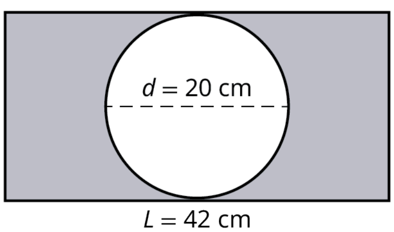
41. Find the area of an octagon if the apothem equals 10 cm and the side length is 12 cm.

42. The area of a circle is $50.3\ \text{in}^2$. What is the radius?

43. You want to install a tinted protective shield on this window. How many square feet do you order?



44. Find the area of the shaded region in the given figure.



Volume and Surface Area

45. Find the surface area of a hexagonal prism with side length 5 cm, apothem 3.1 cm, and height 15 cm.

A triangular prism has an equilateral base and side lengths of 15 in, triangle height of 10 in, and prism length of 25 in.

46. Find the surface area of the triangular prism.

47. Find the volume of the triangular prism.

48. Find the volume of a right cylinder with radius equal to 1.5 cm and height equal to 5 cm.

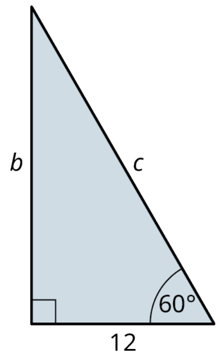
A right cylinder has a radius of 6 cm and a height of 10 cm.

49. Find the surface area of the right cylinder.

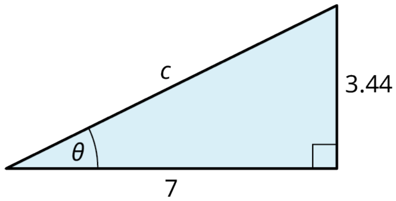
50. Find the volume of the right cylinder.

Right Triangle Trigonometry

51. Find the lengths of the missing sides in the figure shown.

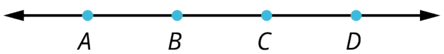


52. Find the measure of the unknown side and angle.



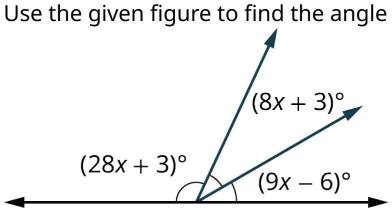
Chapter Test

Use the given figure for the following exercises.

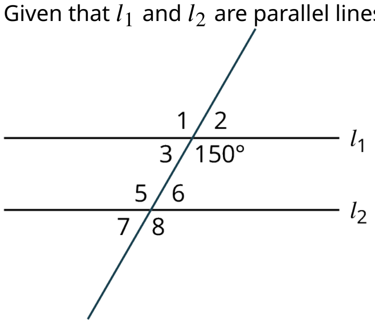


1. Find $\overleftrightarrow{AD} \cap \overline{CD}$.
2. Find $\overline{AB} \cup \overline{BC}$.

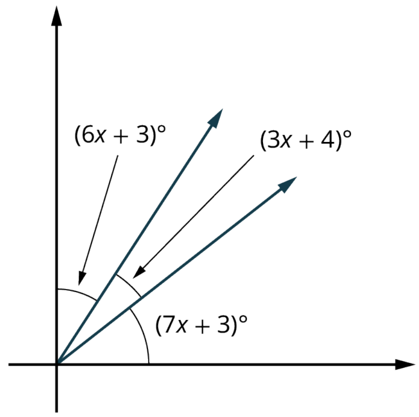
3. Use the given figure to find the angle measurements.



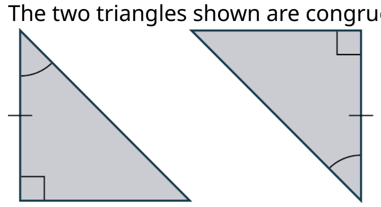
4. Given that l_1 and l_2 are parallel lines, solve the angle measurements for all the angles in the given figure.



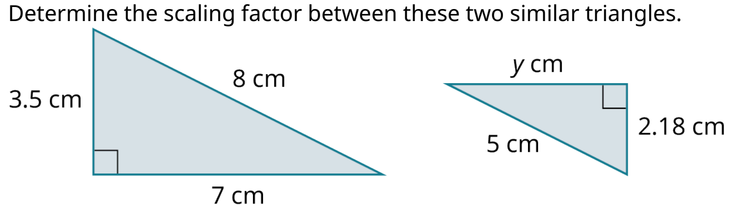
5. Find the angle measurements in the given figure.



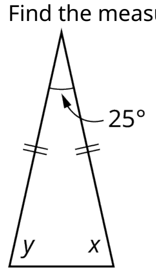
6. The two triangles shown are congruent by what theorem?



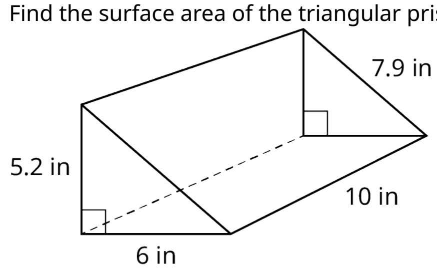
7. Find the sum of the interior angles of a regular heptagon.
8. Determine the scaling factor between these two similar triangles.



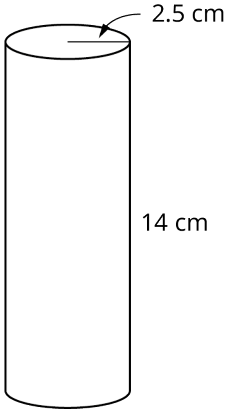
9. Find the measure of the missing angles in the figure shown.



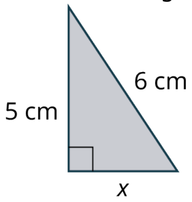
10. Calculate the perimeter of a regular octagon with a side length of 5 cm.
11. Find the measurement of an interior angle of a regular heptagon.
12. Find the sum of the interior angles of a regular pentagon.
13. Find the measure of an exterior angle of a regular pentagon.
14. Find the circumference of the circle with a radius of 3.5 cm.
15. What are the four transformations that are used to produce tessellations?
16. Find the surface area of the triangular prism shown.



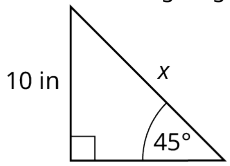
17. Find the volume of the right cylinder in the given figure.



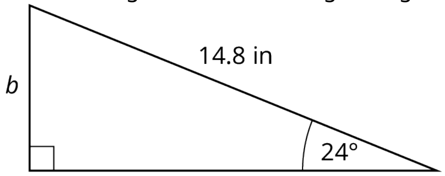
18. Find the missing length in the given figure.



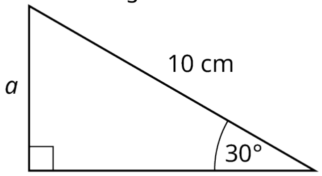
19. Find the missing length in the given figure.



20. Find the length of side b in the given figure.



21. Find the length of side a in the given figure.



22. Find the measure of θ in the given figure.